IN-SILICO DRUG DESIGNING FOR MULTI DRUG RESISTANT *MYCOBACTERIUM TUBERCULOSIS* (MDR-TB) TARGETING ENOYL-ACYL CARRIER PROTEIN REDUCTASE ENZYME (ENOYL-ACP REDUCTASE)

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An project report submitted to The University of Agriculture Peshawar in partial fulfillment of the requirement for the degree of

BACHELOR OF SCIENCE (HONS) BIOINFORMATICS

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ACKNOWLEDGEMENTS

First and foremost, we express our deepest gratitude to Almighty ALLAH for granting us strength, patience, wisdom, and the opportunity to successfully complete this research work.

We would like to convey our sincere and heartfelt appreciation to our respected supervisor, **Dr. Muhammad Tayyab**, for his continuous guidance, valuable suggestions, encouragement, and unwavering support throughout this study. His expertise, constructive feedback, and mentorship played a fundamental role in the successful completion of this research project.

We are also thankful to all our teachers, faculty members, researchers, and laboratory staff for providing us with the necessary facilities, knowledge, and academic support during this work. Their cooperation and guidance greatly contributed to improving the quality of our research.

We would like to acknowledge the support, teamwork, and dedication of all group members whose collective efforts, hard work, and collaboration made this project possible and successful.

TABLE OF CONTENTS

I. INTRODUCTION

1.1 Background

Tuberculosis (TB) is a highly contagious disease caused by the *Mycobacterium tuberculosis*, primarily transmitted through the air (WHO, 2022). The bacterium was first discovered by Robert Koch in 1882, which marked a major breakthrough in medical microbiology and infectious disease research. *M. tuberculosis* primarily infects the lungs, causing pulmonary tuberculosis, but it can also spread to extra pulmonary sites such as the lymph nodes, bones, kidneys, and central nervous system (WHO, 2023) (WHO,25). Tuberculosis spreads through airborne droplets released when infected individuals cough or sneeze. According to WHO Due to its highly contagious nature and global prevalence, TB remains one of the leading causes of death from infectious diseases worldwide. Tuberculosis continues to be a huge peril disease against the human population and according to WHO, tuberculosis is a major killer of the human population after HIV/AIDS. Tuberculosis is highly prevalent among the low socioeconomic section of the population and marginalized sections of the community.(WHO, 2025)

The complete genome sequencing of *Mycobacterium tuberculosis* H37Rv has provided valuable insights into the biology and pathogenicity of this slow-growing bacterium. The genome consists of approximately 4.4 million base pairs and contains nearly 4,000 genes with a high guanine-cytosine (GC) content. A significant portion of the genome is dedicated to lipid metabolism and the synthesis of glycine-rich proteins, which contribute to the survival, virulence, and antigenic variation of *Mycobacterium tuberculosis*,, making them important targets for the development of new therapeutic strategies. (Cole et al., 1998).

The pathogenicity of *Mycobacterium tuberculosis*, is mainly associated with its unique and complex cell wall structure, which is rich in mycolic acids, glycolipids, and arabinogalactan (Garcia-Vilanova et al., 2019). The lipid-rich cell wall provides resistance against environmental stress, host immune defences, and many antimicrobial drugs (Jankute. et al., 2015). The bacterium survives and replicates within host macrophages by evading immune responses, allowing persistent infection and latent tuberculosis. Due to these survival mechanisms, M. tuberculosis remains one of the deadliest human pathogens worldwide.

1.2 Challenges Associated with MDR-TB

According to WHO Tuberculosis (TB) remains one of the most serious infectious diseases worldwide and continues to be a major global health challenge. TB continued to be a significant global health concern, ranking as the second leading cause of infectious disease mortality (Salari et al., 2023). According to the World Health Organization (WHO) approximately 10.7 million people developed TB in 2024, and about 1.23 million deaths were reported globally. Control of tuberculosis (TB) remains one of the most serious challenges to global health (WHO, 2006). Although the greatest numbers of patients live in the highly populous countries of Asia the highest incidence of disease is found in the WHO region of Africa, Nine countries in sub-Saharan Africa have recently reported estimated annual incidences in excess of 600 cases per 100,000 (Corbett et al., 2006). In 2005 there were an estimated 8.8 million new cases and 1.6 million death TB is particularly common in low- and middle-income countries, including Pakistan, India, Indonesia, China, and South Africa, which collectively contribute a major portion of the global TB burden (WHO, 2008).

Another critical challenge is the long duration of TB treatment. Standard therapy requires a combination of antibiotics for at least six months this prolonged treatment often leads to poor patient compliance, which increases the risk of treatment failure and relapse in many cases, incomplete treatment contributes to the development of drug-resistant strains of M. tuberculosis (WHO, 2010). MDR-TB has emerged as a major global health threat due to its complex treatment requirements, long therapy duration, high cost, and poor treatment outcomes compared to drug-susceptible TB. Tuberculosis has become increasingly difficult to control due to the emergence of drug-resistant strains, particularly multidrug-resistant tuberculosis (Shah et al., 2007). According to WHO latest report MDR-TB is defined as tuberculosis caused by *Mycobacterium tuberculosis* strains that are resistant to at least isoniazid and rifampicin, which are the two most effective first-line anti-tuberculosis drugs. This resistance significantly reduces treatment success and increases the complexity, cost, and duration of therapy. The development of MDR-TB is mainly driven by genetic mutations in M. tuberculosis, especially in genes associated with drug targets or drug activation.(WHO, 2024).

Multidrug-resistant tuberculosis (MDR-TB) is caused by Tuberculosis strains that have developed resistance to first-line anti-tuberculosis drugs such as rifampicin and isoniazid. MDR-TB has become a major global challenge for both diagnosis and treatment, with a large proportion of drug regimens showing reduced effectiveness, Several key factors are responsible for the development of multidrug resistance, including improper treatment management by physicians or patients, poor drug penetration into granulomatous lesions due to their complex structure and limited vascularization, intrinsic resistance mechanisms of *Mycobacterium tuberculosis*, survival of non-replicating drug-tolerant bacilli within granulomas, and genetic mutations in M. tuberculosis that represent the primary molecular basis of resistance (Pramod ., 2023). The emergence of MDR-TB has become a major obstacle for global tuberculosis control efforts because it lowers treatment effectiveness, raises healthcare expenses, and leads to increased death rates. Therefore, gaining an accurate understanding of the prevalence and distribution of drug-resistant tuberculosis, especially MDR-TB, is crucial for effective resource allocation, improved disease management, and the prevention of further transmission. (Zager & McNerney, 2008)

Another critical issue is that current diagnostic and therapeutic approaches are often slow, expensive, and not fully effective in identifying all resistant strains. As a result, patients may receive inappropriate treatment, leading to further resistance amplification and transmission within the community. This creates a cycle of persistent infection and increasing disease burden Therefore, MDR-TB represents not only a clinical challenge but also a global health threat that demands rapid diagnostic strategies, novel drug discovery, and computational approaches such as in-silico drug design to identify new inhibitors against essential *M. tuberculosis* targets.

1.3: Aim of the Study:

The main aim of this study is to perform structure-based in-Silico drug designing against Multi-Drug Resistant *Mycobacterium tuberculosis* (MDR-TB) by targeting the Enoyl-Acyl Carrier Protein Reductase enzyme (InhA/Enoyl-ACP Reductase). In this study, the three-dimensional structure of the target protein and subjected to protein preparation and energy minimization for computational analysis. A serious and dangerous problem for tuberculosis (TB) treatment is the development of TB bacteria that no longer respond to the normal medicines used to cure TB. (Lawn & Wilkinson, 2006).

A ligand library were prepared for virtual screening, including Triclosan as a reference ligand for comparative analysis and validation purposes. The screened compounds will then be subjected to molecular docking studies to evaluate their binding affinity and molecular interactions within the active site of the InhA enzyme. Furthermore, Molecular Dynamics (MD) simulation studies were carried out to analyse the stability and dynamic behaviour of the protein–ligand complexes under biological conditions.(Lone at all., 2018)

The overall objective of this research is to identify promising lead compounds that may act as potential inhibitors against MDR *Mycobacterium tuberculosis*, and contribute to the development of novel anti-tuberculosis therapeutics through computational drug discovery approaches.

1.4: Target Selection:

The specific structure used in this Structure-Based In-Silico Drug Designing research project is InhA protein Enoyl-ACP Reductase. The enoyl-ACP reductase from *Mycobacterium tuberculosis*, known as InhA, is a member of an unusual FAS-II system that prefers longer chain fatty acyl substrates for the purpose of synthesizing mycolic acids, a major component of mycobacterial cell walls (Denise et al., 1999). In this crystal structure inhA provides critical structural and functional insights into the enzyme Enoyl-ACP Reductase (InhA), which is essential for the survival of *Mycobacterium tuberculosis* (Chollet et al., 2015). Enoyl-ACP reductase plays a central role in the fatty acid synthesis II (FAS-II) pathway, catalysing the NADH to reduce the trans double bond between positions C2 and C3 of a fatty acyl chain linked to the acyl carrier protein (Rozwarski et al., 1999). InhA is considered a promising molecular target for the development of novel anti tubercular drugs (Aurélien et al., 2025). The high-resolution structure of 4TRO enables detailed visualization of the enzyme’s active site, ligand-binding interactions, and catalytic mechanism, making it highly valuable for understanding enzyme function and for structure-based drug design.

Protein Data Bank ID structure 4TRO was selected for this research study because it represents the crystal structure of the wild-type InhA (enoyl-ACP reductase) enzyme of Tuberculosis, which plays a critical role in the biosynthesis of mycolic acids, essential components of the mycobacterial cell wall. The 4TRO structure was provides the first crystal structure of the apo form of wild-type InhA at 1.80 Å resolution, along with the structure of InhA complexed with the synthetic metabolite of the anti-tubercular drug Isoniazid refined to 1.40 Å resolution. The study demonstrated that this metabolite can displace and replace the co-factor NADH within the enzyme active site, providing important structural insights into ligand binding and inhibition mechanisms. Therefore, the 4TRO structure was considered highly suitable for molecular docking and structure-based drug design studies targeting InhA. Enoyl-ACP reductases participate in fatty acid bio synthesis by utilizing NADH to reduce the trans double bond between positions C2 and C3 of a fatty acyl chain linked to the acyl carrier protein. The enoyl-ACP reductase from *Mycobacterium tuberculosis*,, known as InhA, is a member of an unusual FAS-II system that prefers longer chain fatty acyl substrates for the purpose of synthesizing mycolic acids, a major component of mycobacterial cell walls (Denise et al., 1999). The enoyl-thioester reductase InhA catalyzes an essential step in fatty acid biosynthesis of *Mycobacterium tuberculosis* and is a key target of anti-tuberculosis drugs to combat multidrug resistant *M. tuberculosis* strains this has prompted intense interest in the mechanism and intermediates of the InhA reaction (Bastian et al., 2018). InhA, the enoyl acyl carrier protein reductase (ENR) from *Mycobacterium tuberculosis*, is one of the key enzymes involved in the type II fatty acid biosynthesis pathway of M. tuberculosis. Inhibition of InhA disrupts the biosynthesis of the mycolic acids that are central constituents of the mycobacterial cell wall. (Xin et al., 2007). Mycobacterial enoyl-ACP-reductase, an enzyme contributing in mycolic acids biosynthesis, has been established as promising target of novel ant mycobacterial drugs. The development of inhibitors active without previous activation by catalase/peroxidase system (e.g. isoniazid), seems to be rational approach. (Holas et al., 2014)

II. REVIEW OF LITERATURE

Rasul et al. (2025) reported that the drug development process remains one of the most critical challenges in the pharmaceutical industry due to its time-consuming, costly, and complex nature, particularly in the discovery of novel therapeutic agents for various diseases. One of the earliest and most crucial stages in this process is drug target identification, which often requires substantial time and resources. Although traditional experimental approaches such as *in vivo* and *in vitro* methods are considered reliable, they are limited in their ability to efficiently analyze large-scale biological and chemical data. As a result, these methods can be slow, resource-intensive, and may lead to inefficient outcomes in the early stages of drug discovery.

Kuppuswamy et al. (2022) investigated that the emergence of mutated drug-resistant strains of *Mycobacterium tuberculosis* has significantly intensified the need for the development of more effective chemotherapeutic strategies against multidrug-resistant tuberculosis (MDR-TB). In this context, *Mycobacterium tuberculosis* enoyl-acyl carrier protein reductase (InhA), a key enzyme involved in the fatty acid elongation system (FAS-II pathway), has gained considerable attention as a promising molecular target for drug discovery. InhA plays a crucial role in mycolic acid biosynthesis, which is essential for the survival and virulence of the bacterium. Due to its essential biological function and validated role in existing anti-tuberculosis drug mechanisms, InhA has been widely selected as a potential target for the rational design of novel anti-TB agents.

Singh et al. (2022) demonstrated this Tuberculosis enoyl-acyl carrier protein reductase (InhA) has been validated as an important therapeutic target for the treatment of tuberculosis and is considered a promising enzyme for anti-tubercular drug discovery. InhA plays a crucial role in the type II fatty acid biosynthesis (FAS-II) pathway, which is responsible for the synthesis of mycolic acids essential for the formation and maintenance of the mycobacterial cell envelope. Due to its essential biological function, inhibition of InhA has emerged as an effective strategy for controlling the growth and survival of *Mycobacterium tuberculosis*.

Phusi et al. (2023) reported that 2-trans enoyl-acyl carrier protein reductase (InhA) has been widely recognized as a promising molecular target for the development of novel chemotherapeutic agents against *Mycobacterium tuberculosis*, particularly in the context of drug-resistant tuberculosis. Numerous experimental studies have investigated the inhibitory effects of different chemical scaffolds on InhA activity; however, the observed variation in inhibitory potency among structurally similar compounds could not be fully explained solely on the basis of their physicochemical properties. To address this limitation, molecular simulation approaches have been increasingly employed to provide deeper insights into ligand–target interactions at the atomic level.

Halder & Elma (2021) demonstrated that computational biology and bioinformatics approaches provide an efficient platform for the discovery of novel anti-tuberculosis drugs. Through virtual screening, molecular docking, ADMET analysis, and molecular dynamics simulations, the authors identified several compounds with strong binding affinity toward the InhA and EthR proteins of *Mycobacterium tuberculosis*. Among the screened molecules, compounds C22 and C29 exhibited the most favorable pharmacokinetic properties, stable protein–ligand interactions, and superior binding energies compared with some existing anti-tuberculosis drugs. The study highlighted that computational methods can rapidly screen a large number of compounds, reduce the cost and time associated with traditional drug discovery, minimize the risk of adverse drug effects, and predict potential drug resistance. These findings suggest that in-silico approaches are valuable tools for identifying promising lead compounds for further experimental validation and anti-tuberculosis drug development

Almihyawi et al. (2022) demonstrated the crystal structure of the EPSP synthase enzyme was retrieved from the Protein Data Bank (PDB) database. The structure was resolved at a resolution of 2.37 Å with an R-value of 0.186 and is available under PDB ID: 5BUF. Prior to molecular docking studies, the protein structure was carefully prepared by removing extraneous chains, water molecules, and co-crystallized ligands using UCSF Chimera software (version 1.15). Subsequently, the enzyme structure was subjected to energy minimization to relieve steric clashes and optimize geometry, employing 1000 steps of steepest descent followed by conjugate gradient algorithms with a step size of 0.02 Å. The AMBER ff14SB force field was applied to parameterize the protein backbone and side-chain residues. To evaluate the structural quality and stereochemical stability of the minimized model, a Ramachandran plot was generated using PDBsum and compared with the unminimized structure. The superimposed structures of the energy-minimized and experimentally determined EPSP synthase enzyme demonstrating structural consistency after minimization.

Almihyawi et al. (2022) identified potential natural inhibitors against EPSP synthase enzyme, the CMNPD database (https://www.cmnpd.org/) was utilized as a primary source of marine-derived bioactive compounds. This manually curated and freely accessible database is considered a valuable repository of novel lead-like molecules obtained from diverse marine organisms, including bacteria, fungi, and algae, and contains approximately 47,000 structurally diverse compounds. The dataset was downloaded in SDF format and subsequently subjected to systematic filtering using the FAFDrugs4 online server. During his study process, drug-likeness criteria, toxicophore filters, and Eli Lilly MedChem rules were applied in a stepwise manner to eliminate non-drug-like, toxic, and PAINS (Pan-Assay Interference Compounds) structures. Following filtration, the selected compounds were imported into PyRx (version 0.8), where they were energy-minimized and converted into PDBQT format to prepare them for molecular docking studies.

Almihyawi et al. (2022) showed that for Structure-based virtual screening (SBVS) of the filtered compound library against EPSP synthase enzyme was performed using two widely used molecular docking platforms, namely Genetic Optimization for Ligand Docking (GOLD) 5.2 and PyRx integrated with AutoDock Vina. This dual-docking strategy was employed to validate docking predictions and improve the reliability of binding affinity results through consensus scoring. In both docking approaches, a rigid receptor docking protocol was applied to maintain the structural integrity of the target protein during simulations. The resulting protein–ligand complexes were analyzed comparatively, and the top three hits were selected based on the highest GOLD fitness scores and the lowest binding energy values obtained from AutoDock Vina (kcal/mol). The best-docked ligand–enzyme complexes were retrieved in PDB format and further visualized and analyzed using UCSF Chimera (version 1.15) and BIOVIA Discovery Studio Visualizer.

III. MATERIALS AND METHODS

This research work were carried out at the institute of Biotechnology and Genetic Engineering (IBGE) at the faculty of molecular and biological sciences, The university of Agriculture Peshawar.

Flowchart for Methodology

Target Selection (InhA, PDB ID: 4TRO)

↓

Protein Preparation

↓

Protein Refinement & Validation

↓

Active Site Prediction (CASTp)

↓

library preparation / Ligand Retrieval (CMNPD)

↓

Drug-Likeness & ADMET Screening

↓

hERG Toxicity Screening

↓

Molecular Docking (PyRx/AutoDock Vina)

↓

Interaction Analysis

↓

Molecular Dynamics Simulation

↓

Lead Compound Identification
(Acetyllaurefucin)

1. **Target Selection**
   - InhA (Enoyl-ACP Reductase)
   - PDB ID: 4TRO

2. **Protein Preparation**
   - Structure retrieval
   - Preparation (hydrogen addition, Removing hatatoms, charge assignment)
   - Energy minimization (Chimera/AMBER)
   - Stability Analysis/Validation (Ramachandran plot)

3. **Active Site Prediction**
   - CASTp analysis
   - Active site identification
   - Grid box generation

4. **Liberary Preparation/ ligand selection**
   - CMNPD database retrieval (500 compounds
   - Energy minimization (Chimera)
   - Drug-likeness screening (SwissADME)
   - ADMET filtering (including hERG)
   - Selection of 7 compounds

5. **Molecular Docking**
   - PyRx/AutoDock Vina
   - Validation with reference ligands (NADH, Triclosan)
   - Virtual screening of 7 compounds
   - Selection of top compound (acetyllaurefucin)

6. **Molecular Dynamics Simulation**
   - GROMACS (100 ns, triplicate)
   - RMSD, RMSF, Rg, hydrogen bond analysis
   - MM-PBSA binding free energy calculation

IV. RESULTS

4.1: Structure Retrieval of Enoyl-ACP Reductase:

First of all, the Protein Crystal structure of Enoyl-ACP reductase enzyme was retrieved from Protein Data Bank (PDB). The structure is determined at a resolution of 1.40 Å. R-value of 0.123 and submitted under the PDB ID of 4TRO.

Structural Details.
Protein Name: Enoyl-ACP Reductase (InhA)
Organism: Mycobacterium tuberculosis
PDB ID: 4TRO
Format: PDB format
Method of Determination: X-ray crystallography
Resolution: 1.40 Å
Cofactor: NADH/NAD+ (commonly present in InhA structures)
Structure Type: Protein-ligand complex

The selected structure has a high resolution of 1.40 Å, which indicates a very accurate and reliable atomic model. In structural biology, lower resolution values correspond to higher structural clarity, making this structure highly suitable for molecular docking and virtual screening studies**.** Studies have emphasized that high-resolution crystal structures of InhA, such as 4TRO, are essential for designing new inhibitors against multidrug-resistant tuberculosis (MDR-TB) (Pan & Tonge, 2012).

4.2: Protein preparation:

The crystal structure of Enoyl-ACP Reductase (InhA) was prepared using Discovery Studio Visualizer to make it suitable for molecular docking studies.

The protein structure were opened in Discovery Studio Visualizer. Then, the structure was cleaned by removing unnecessary molecules such as water molecules and heteroatoms (HETATM). This was done by going to the View menu, selecting the Hierarchy option, and identifying these molecules, which were then deleted. Water molecules were removed during protein preparation because many crystallographic water molecules are not directly involved in ligand binding and may interfere with docking calculations by occupying the active site.. Removing them helps in avoiding unnecessary interactions and improves docking accuracy. And hetero-atoms include co-crystallized ligands, ions, or non-essential molecules. These can block the active site or affect docking, so they are removed unless specifically required for analysis.After cleaning, hydrogen atoms were added using the Chemistry section by selecting the Add Hydrogen option. This step is important because most protein structures 4TRO was X-ray crystallography and some X-ray crystallography structures do not include hydrogen atoms. Adding hydrogen helps to complete the structure, improve bonding, and allow proper interactions during docking. Hydrogen atoms were added because X-ray crystal structures often lack visible hydrogen atoms. Their addition improves charge distribution, hydrogen bonding, and docking accuracy.

- Completes valency
- Improves charge distribution
- Enables proper hydrogen bonding
- Enhances docking accuracy

Finally, the prepared protein structure was saved in PDB format for further analysis, which is shown in figure 4.3 for further computational analysis.

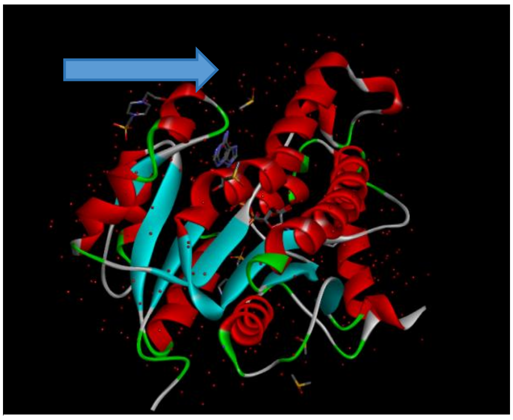

Figure: 4.1 Before minimization there are water Molecules and heteroatoms are showing.

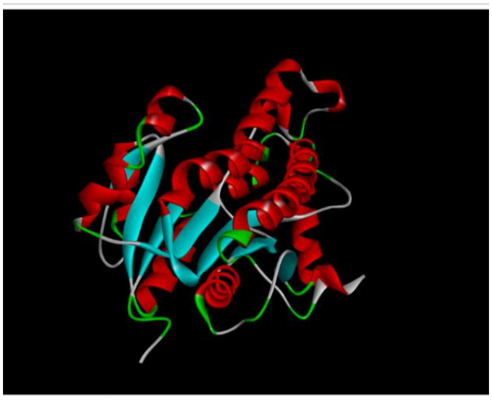

Figure: 4.2 After Removing Water molecules and *HETATMs*.

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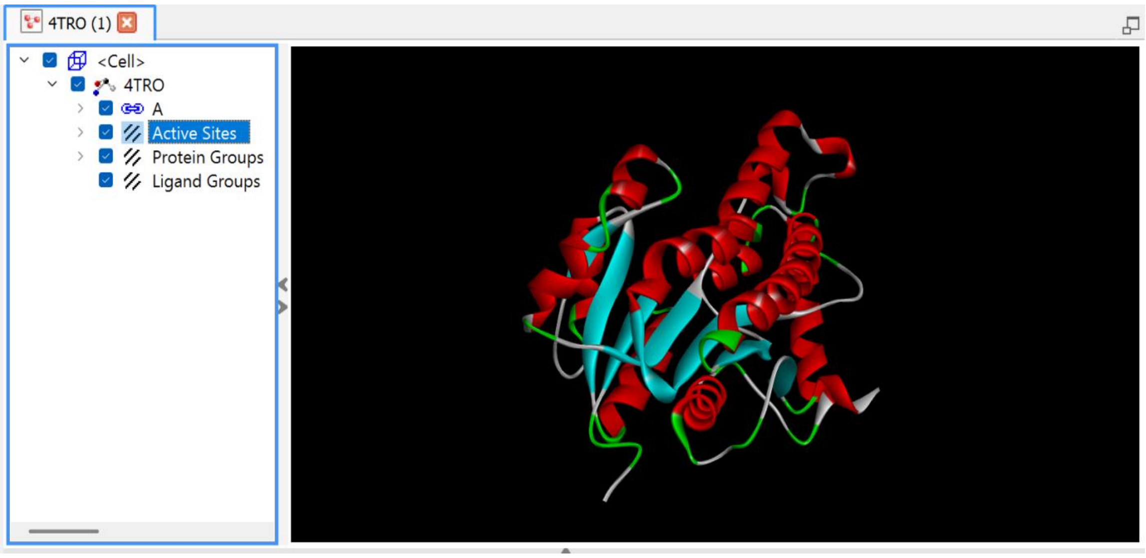

Figure 4.3: After minimization, water molecules and heteroatoms were removed to ensure structural clarity, and hydrogen atoms were added to stabilize the protein structure.

4.3: Energy Minimization:

The prepared structure of InhA was further refined using the online tool Galaxy Refine 2 web server (https://galaxy.seoklab.org/cgi-bin/submit.cgi?type=REFINE) to improve its structural quality and stability for molecular docking studies.

After completing the protein preparation step (removal of water molecules and hetero-atoms, and addition of hydrogen atoms) the cleaned protein structure was submitted to the Galaxy Refine server for minimization where performs (Filling missing Residues, | Mild Relaxation, | and Aggressive Relaxation). The refinement process was performed to correct structural imperfections and optimize the geometry of the protein. Galaxy Refine improves the protein structure through several processes.After refinement Galaxy Refine generated multiple refined models and displayed the results in a table format shown in (Figure 4.4). This included important evaluation parameters such as structural quality scores and energy values. From these results, the best performing model 1 was selected based on improved structural parameters.

Model 1 in the (figure 4.4) shown in the table was selected and download in PDB format because it showed the best structural quality and stability among all refined models showing in the table.

View in PV [Model 1] [Model 2] [Model 3] [Model 4] [Model 5] [All Models]

Download [Model 1] [Model 2] [Model 3] [Model 4] [Model 5] [All Models]

Structure Information

Model	GDT-HA	RMSD	MolProbity	Clash score	Poor rotamers	Rama favored
Initial	1.0000	0.000	1.507	6.1	0.5	97.0
MODEL 1	0.9963	0.244	1.714	9.3	2.0	98.1
MODEL 2	0.9963	0.225	1.471	7.6	0.5	97.7
MODEL 3	0.9972	0.243	1.390	7.1	1.0	98.1
MODEL 4	0.9953	0.248	1.363	6.6	0.5	98.1
MODEL 5	0.9981	0.226	1.438	8.1	0.0	98.1

Figure 4.4 : Minimized Results were obtained through filling missing residues, followed by mild and aggressive relaxation to stabilize the protein structure. Using Galaxy Refine.

- GDT-HA (Global Distance Test – High Accuracy) is higher because close to 1.
- RMSD (Root Mean Square Deviation) is very good 0.244Å is stable structure.
- MolProbity is very good because around 1–2 high quality structure.
- Clash Score is low.
- Poor Rotamers mean side-chain conformations of amino acids. so it is better.
- Rama Favored (Ramachandran Favored %) is the Percentage of residues in allowed regions of Ramachandran plot. so 98.1 % is very good minimized structure.

4.4: Stability Analysis

After completing the protein minimization of the InhA protein using Galaxy Refine2 online server the refined structure was submitted to the PDBsum online tool (https://www.ebi.ac.uk/thornton-srv/databases/pdbsum/Generate.html) to generate a Ramachandran plot and statistical analysis of InhA protein. The Ramachandran plot is used to analyse the backbone dihedral angles (φ and ψ) of amino acid residues and to determine the quality and stability of a InhA protein structure. After submitting my target protein into PDBsum they gave me a plot showing in (Figure 4.4).

The Ramachandran plot displays the distribution of phi (φ) and psi (ψ) angles of the protein backbone. in the plot show us in (figure 4.5, 4.6). These are the statistics how many amino acids in targeted protein are positioned in favourable regions of the Ramachandran plot. They tell us about the overall quality and stability of your protein backbone geometry.

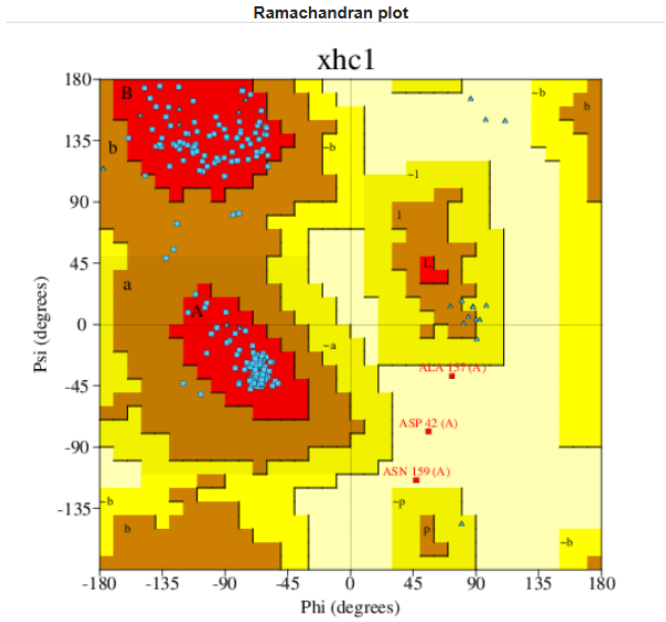

Figure 4.5: Ramachandran plot show us the stability of our target protein using PDBsum Online software.

In Red: Most favoured regions (best positions for alpha-helix and beta-sheet).

In Brown: Additional allowed regions (acceptable).

In Yellow: Generously allowed regions (less common but okay).

In White: Disallowed regions (unfavourable - atoms clash).

1. Ramachandran Plot statistics

```
                                             No. of
                                            residues     %-tage
                                             ------      ------
Most favoured regions      [A,B,L]             212        94.2%
Additional allowed regions [a,b,l,p]            10         4.4%
Generously allowed regions [~a,~b,~l,~p]         0         0.0%
Disallowed regions         [XX]                  3         1.3%*
                                              ----       ------
Non-glycine and non-proline residues           225       100.0%

End-residues (excl. Gly and Pro)                 2

Glycine residues                                28
Proline residues                                13
                                              ----
Total number of residues                       268
```

Based on an analysis of **118** structures of resolution of at least **2.0** Angstroms and *R*-factor no greater than **20.0** a good quality model would be expected to have over **90%** in the most favoured regions [A,B,L].

2. G-Factors

```
                                                 Average
Parameter                             Score       Score
---------                             -----       -----
Dihedral angles:-
     Phi-psi distribution              0.22
     Chi1-chi2 distribution            0.64
     Chi1 only                         0.34
     Chi3 & chi4                       0.79
     Omega                             0.48
                                                   0.44
                                                  =====
Main-chain covalent forces:-
     Main-chain bond lengths          -0.09
     Main-chain bond angles           -0.18
                                                  -0.14
                                                  =====

     OVERALL AVERAGE                               0.21
                                                  =====
```

G-factors provide a measure of how **unusual**, or out-of-the-ordinary, a property is.

```
Values below -0.5*  - unusual
Values below -1.0** - highly unusual
```

Important note: The main-chain bond-lengths and bond angles are compared with the Engh & Huber (1991) ideal values derived from small-molecule data. Therefore, structures refined using different restraints may show apparently large deviations from normality.

Figure 4.6: Ramachandran plot statistical analysis of our protein.

In the given Ramachandran plot Statistics they show us about our target protein.Most Favoured Regions: 212 residues (94.2%) is excellent, because a good-quality protein model should have >90% residues in this region. Additionally Allowed Regions: 10 residues (4.4%) is normal and acceptable. Generously Allowed Regions: 0 residues (0.0%) is excellent because there is no residues in weakly allowed regions. Disallowed Regions: 3 residues (1.3%) normal because a small percentage is acceptable, but ideally should be minimized.

Glycine residues: 28 (flexible, often appear in unusual regions). Proline residues: 13 (restricted angles due to rigid structure).Total residues: 268. The results showed that 94.2% of residues were located in the most favoured regions, exceeding the standard threshold of 90% for a high-quality model. Additionally, 4.4% of residues were found in allowed regions, while only 1.3% were present in disallowed regions, indicating minimal structural deviations. Therefore our targeted protein model is reliable and suitable for molecular docking and in-silico drug design studies.

4.5: Active Sites Prediction:

A protein (enzyme) active sites is the specific three-dimensional region where a substrate or ligand binds and a biochemical reaction occurs. It is formed by a group of amino acid residues that create a pocket or cavity with a unique shape and chemical environment. For active site prediction of the target protein (inhA) was performed using the Computed Atlas of Surface Topography of proteins (CASTp) server. CASTp employs computational geometry-based algorithms, particularly alpha shape theory and Delaunay triangulation, to accurately identify surface pockets and internal cavities of the protein structure. The method utilizes a probe sphere to explore the solvent-accessible surface and detect potential ligand-binding regions. It provides precise quantitative measurements, including pocket area and volume, which are essential for identifying biologically relevant active sites. The largest and most significant pocket, often corresponding to the substrate or cofactor binding region, was identified and noticed with protein for further molecular docking analysis.

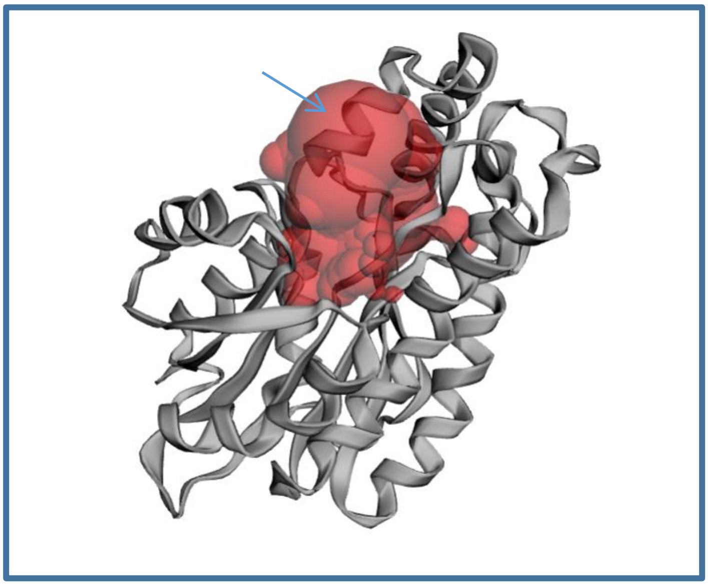

Figure 4.7: The identified surface pockets and internal cavities within the protein structure using CASTp online server the red highlight region show us effective binding sites .

The predicted active site pocket is composed of amino acid residues from Chain A (Figure. 4.7), with the sequence annotation indicating specific residue types lining the binding cavity.

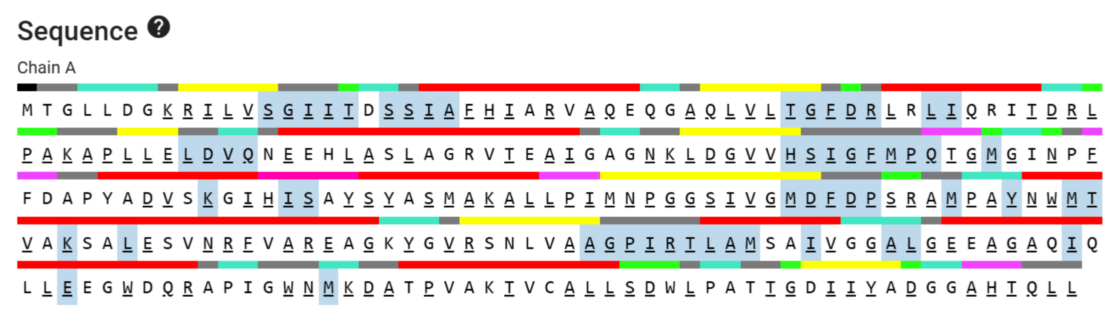

Figure 4.8: Identified CASTp tool results of prominent surface pocket with significant Volume and surface area.

Red regions: Likely α-helices or catalytic residues. Yellow regions: Possible β-sheets or structural elements. Cyan/turquoise: Loops or connector regions.Green/Magenta: Alternative structural designations.Grey: Other structural features.

4.6: Grid Box Setting:

After identification and prediction of the active site region of the target protein, a grid box was defined around the selected binding pocket for molecular docking studies. The grid box represents a three-dimensional search space. It ensures that the docking simulation is focused on the active site or predicted binding pocket of the protein rather than exploring the entire protein surface. grid box is to improve the accuracy, efficiency, and biological relevance of docking results. to ensure accurate targeting of the functional binding pocket in this study the grid box was carefully positioned over the predicted active site region (figure 9) with centre coordinates. to ensure accurate targeting of the functional binding pocket.The InhA protein structure was loaded into AutoDock software and oriented according to the binding site information predicted by CASTp analysis. The active site residues were identified, and the region corresponding to the predicted binding pocket was visually examined, which was highlighted as the active (red) region in the CASTp results.

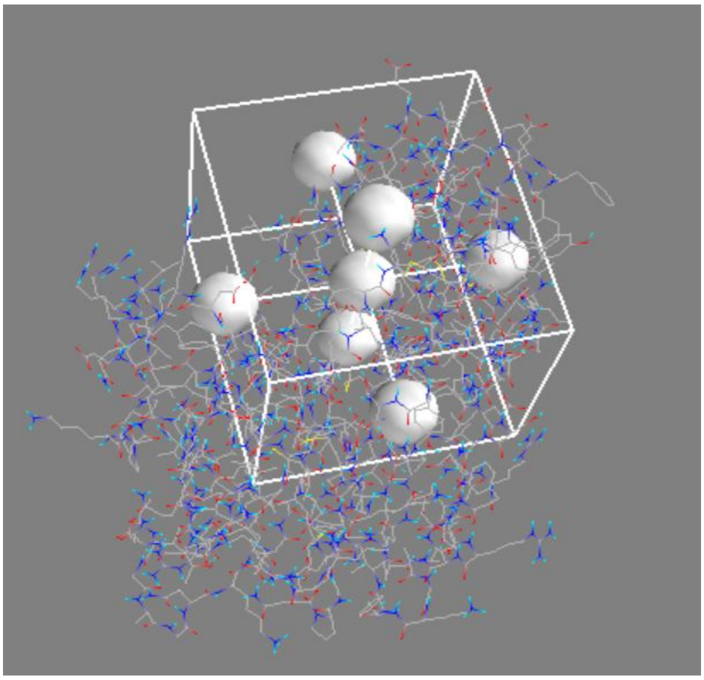

Figure 4.9: Grid box setting to ensure accurate targeting of the functional binding pocket.

4.7 Library preparation of compounds:

In this study, molecular docking was performed against the InhA enzyme of *Mycobacterium tuberculosis*." against the InhA (enoyl-acyl carrier protein reductase**)** enzyme of *Mycobacterium tuberculosis*, The enzyme is essential for mycolic acid biosynthesis, making it an important drug target for anti-tubercular drug discovery.In this research study we used one reference compound/ligand were used as a standard for validation and comparison of docking results and 500 compounds were testing and best matched five to ten tested compounds were selected or designed and prepared by us for virtual screening and further analysis.The reference ligand were Triclosan. Because Triclosan is a well-known direct inhibitor of the InhA enzyme involved in the fatty acid biosynthesis pathway of *Mycobacterium tuberculosis*. It inhibits mycolic acid production, leading to disruption of the bacterial cell wall. Although triclosan itself is not used clinically due to toxicity concerns, it serves as an important scaffold for the development of novel anti-tubercular agents. Triclosan is a potent inhibitor of the InhA enzyme, an essential enoyl reductase in *Mycobacterium tuberculosis*. The compound exhibited strong binding affinity with inhibition constants ($Ki' \approx 0.2$ μM), indicating effective enzyme inhibition. Structural analysis revealed that the Y158 residue plays a critical role in ligand binding. Furthermore, mutations such as M161V and A124V reduced triclosan sensitivity, suggesting a mechanism for resistance development. Importantly, certain isoniazid-resistant mutations (I47T and I21V) remained susceptible to triclosan, highlighting the potential of triclosan-based inhibitors in overcoming drug-resistant tuberculosis. These findings support the use of InhA as a promising target for structure-based drug design. (Parikh et al., 2000).

To identify novel potential inhibitors against the InhA enzyme of *Mycobacterium tuberculosis*, a diverse natural compound library was retrieved from the CMNPD where Plantae-derived natural products were selected with molecular weights between 100–500 g/mol. Preference was given to chemically diverse and drug-like molecules to enhance screening coverage. Those compound are selected with available three-dimensional structures were included to facilitate molecular docking studies. 500 compounds were retrieved from the CMNPD database. For further processing and preparing for virtual screening and molecular docking analysis.

4.8 Drug-likeness screening

For Drug-likeness screening we use online tool SwissADME web server (http ://www.swissadme.ch, accessed January 2025) Total of 500 compounds were uploaded to SwissADME. The physicochemical and pharmacokinetic properties of all compound s were analyzed, and the
results were obtained in the form of graphical and tabular outputs. One of shown in (fig ure 4.10).

After performing drug-likeness and ADME screening using the SwissADME web server, all compounds in the prepared library were carefully evaluated based on key parameters, including physicochemical properties, lipophilicity, water solubility, pharmacokinetics, drug-likeness rules, and medicinal chemistry alerts Shows one of result of them in the tables. And 50 compound are selected for further analysis to do molecular docking on the basis of physicochemical properties, lipophilicity, water solubility, pharmacokinetics, drug-likeness.

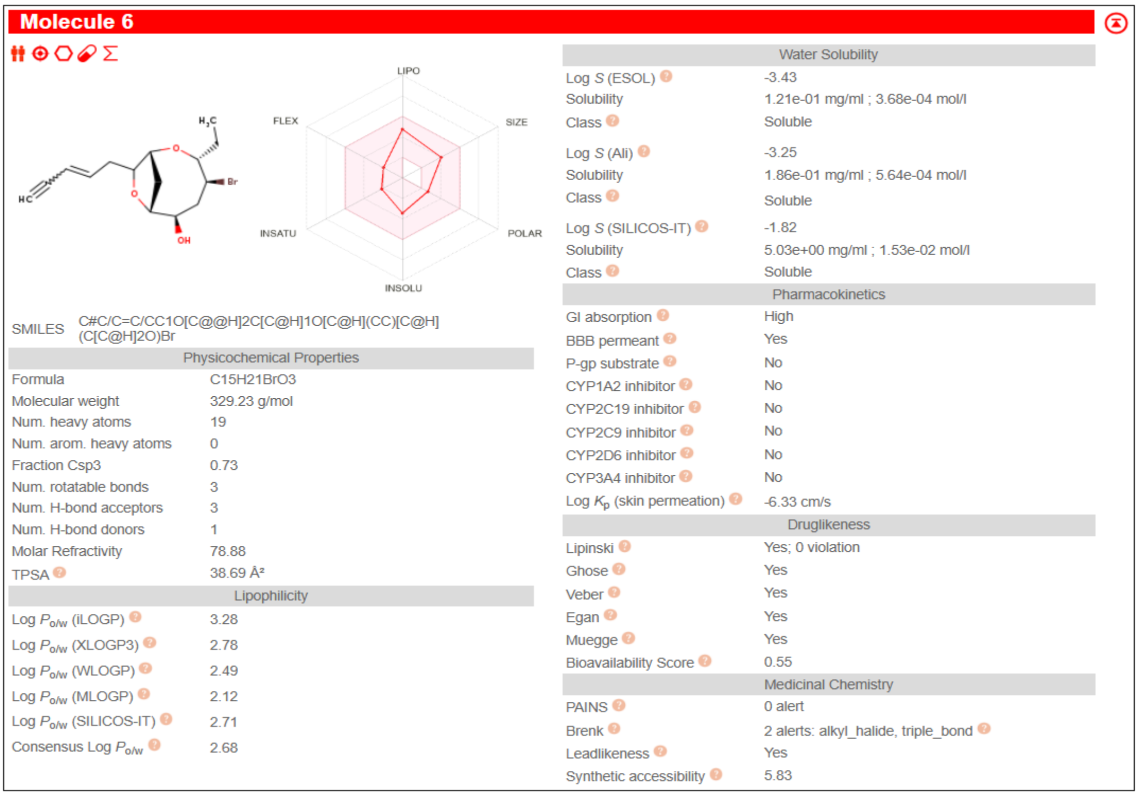

Figure 4.10: The physicochemical and pharmacokinetic properties one of the selected prepared compound

Table No 4.1: Physicohemical properties of our targeted ligand compounds.

Parameter	Result	Normal State	Meaning
Molecular Weight	329.23 g/mol	< 500 g/mol	Good - lighter molecules pass through membranes easier
H-bond Donors	1	≤ 5	Good - few hydrogen bond donors (atoms that share electrons)
H-bond Acceptors	3	≤ 10	Good - limited acceptors means better cell absorption
Rotatable Bonds	3	≤ 10	Good - molecule is somewhat rigid (flexible molecules are harder to absorb)
Topological Surface Area (TPSA	38.69 Å²	< 140 Å²	Good - measure of polarity; affects how easily drug enters cells
Fraction Csp3	0.73>	0.25	Good - indicates saturated carbons (shows drug-like character)

Table no 4.2: Lipophilicity of our targeted ligands compounds.

Parameter	Result	Normal State	Meaning
Consensus Log P	2.680	0 - 5	Good - balanced lipophilicity.
Log P (ILOGP)	3.28	-	Slightly higher (moderate lipophilicity)
Log P (XLOGP3)	2.78	-	Good balance
Log P (WLOGP)	2.49	-	Good balance

Table No 4.3: Water solubility of our targeted ligands compounds.

Method	Log S Value	Solubility	Class	Status
ESOL	-3.43	0.121 mg/ml	Soluble	Good
Ali	-3.25	0186 mg/ml	Soluble	Good
SILICOS-IT	1.82	5.03 mg/ml	Soluble	Excellent

Table No 4.4: Pharmacokinetics properties of our targeted ligands compounds.

Parameter	Result	Normal State	Meaning
GI Absorption	HIGH	High = Good	Our drug easily enters the bloodstream from stomach.
BBB Permeant	YES	Yes/No depends on target	Can cross blood-brain barrier (reach brain tissues).
P-gp Substrate	no	Better	Not pumped out of cells by efflux proteins.
CYP Inhibitors	All No	No is Excellent	Won't interfere with liver metabolism of other drugs (safe combination therapy)
Skin Permeation	-6.33 log kp (cm/s)	< -2.5 cm/s ✗	Poor - unlikely to penetrate skin (OK if oral/injection route).

Table No 4.5: Drug- Likeness of our targeted ligands compounds

Rule	Result	Passes?	Means
Lipinski	0 violations	YES Perfect	passes classic drug criteria
Ghose	YES	YES	Passes additional filtering rules
Veber	YES	YES	Passes flexibility rules
Egan	YES	YES	Passes cell permeability rules
Muegge	YES	YES	Passes medicinal chemistry rules
Bioavailability Score	0.55	MODERATE	0.55% bioavailability (good, not excellent)

Table No 4.6: Medicinal chemistry of our targeted ligands compounds.

Parameter	Result	Normal State	Meaning
PAINS Alerts	0	0 = Excellent	No problem structures (known troublemakers)
Brenk Alerts	2 alerts	-	2 issues found: 1) Alkyl halide (C-Br bond) 2) Triple bond
Leadlikeness	Yes	YES	Good candidate for further development
Synthetic Accessibility	5.83	< 6 = Easy	Moderately easy to synthesize in laboratory

Based on these standard criteria's and normal acceptable ranges, the top 50 best-matched molecules were selected from the initial dataset. These compounds showed favourable drug-like characteristics and minimal violations across all evaluated parameters. And these 50 compounds are finalized for further analysis before docking.

4.9 hERG Prediction Analysis:

hERG protein is a voltage-gated potassium ion channel located in cardiac cells. This protein plays a crucial role in regulating the electrical activity of the heart by controlling the rapid delayed rectifier potassium current (IKr). The primary function of the hERG channel is to facilitate the repolarization phase of the cardiac action potential. During this phase, potassium ions flow out of the cardiac cells, helping restore the resting membrane potential after each heartbeat. Proper functioning of this channel is essential for maintaining a normal heart rhythm. hERG is considered a critical safety target in drug discovery because its inhibition can lead to severe cardiotoxic effects. Many compounds unintentionally block the hERG channel, which may result in

Prolongation of the QT interval. Cardiac arrhythmias. Life-threatening conditions such as torsade de pointes. Due to these risks, regulatory authorities require early evaluation of hERG inhibition during drug development. Therefore, compounds that show strong hERG blocking activity are usually eliminated in the early stages of screening. Assessing hERG liability helps in selecting safer drug candidates and reducing the chances of late-stage failure.

In the present research study the analysis was performed using the Pred-hERG online tool.(https://predherg.labmol.com.br/) The target ligand/compound was submitted to the Pred-hERG server, and the prediction results indicated that it is a non-blocker of the hERG potassium channel. This suggests that the compound does not significantly inhibit the hERG channel and is less likely to cause cardiotoxic effects such as QT interval prolongation or cardiac arrhythmias. Based on the obtained results, only compounds predicted as non-blockers were considered safe and selected for further analysis. From the screened dataset, a total of 7 compounds demonstrated non-blocker behaviour with low risk of cardiotoxicity. Therefore these compounds shortlisted for subsequent molecular docking studies to evaluate their binding affinity and interaction with the target protein.

4.10 Molecular docking:

Molecular docking was performed using the PyRx platform, which integrates AutoDock Vina for predicting ligand–protein interactions. A total of seven newly selected ligands(Triclosan” neoconcinndiol hydroperoxide” 1,1,3-tribromoacetone” laurefucin” acetyllaurefucin” CMNPD357” CMNPD358” ochtodiol) are loaded which we have prepared for docking, and one reference ligand Triclosan are loaded using as a reference ligand for comparative analysis.

Both the target protein (macromolecule) and all seven ligands were imported into PyRx software. Prior to docking, ligand preparation was carried out through normalization, which included correction of protonation states, assignment of proper charges, and energy minimization to ensure structural stability and accuracy. After preparation, all ligands were converted into the required PDBQT format for compatibility with the docking engine.

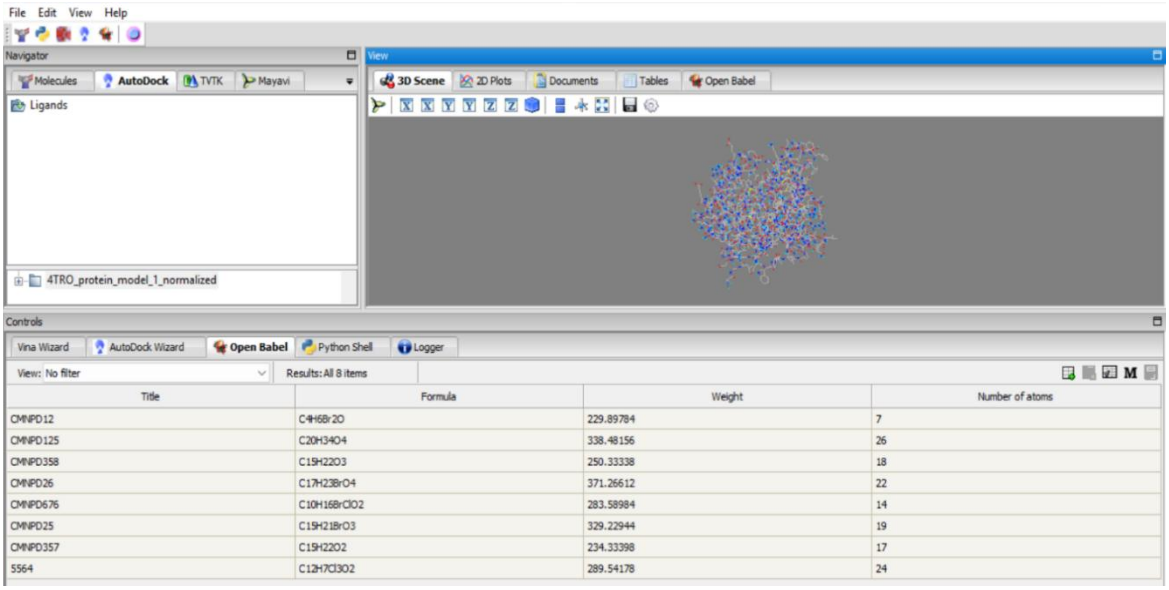

Figure 4.11: Both the target protein (macromolecule) and the seven ligands were imported into PyRx software after preparation for molecular docking analysis)

Following ligand preparation, all compounds were selected for docking. And then grid box was defined over the active site of the protein (showing in figure: 4.11) which had been previously identified in the 'Active Site Prediction and Grid Box Setting' step. The grid box parameters were set as follows:

Center coordinates: X = 3.314 Å, Y = –27.64 Å, Z = 12.36 Å

Dimensions: X = 60 Å, Y = 76 Å, Z = 50 Å

These parameters ensured that the docking process was focused specifically on the predicted binding pocket of the target protein.

After setting the grid box, molecular docking was carried out using the AutoDock Vina engine integrated within PyRx. The docking process was completed successfully, generating a total of 73 binding conformations. After analysing and from these results, the binding pose of the reference ligand (Triclosan) and the best top-ranked ligand-protein complex with the lowest binding energy were selected and downloaded in PDB format.

From the docking results, the complex at rank 2 is result of reference ligand and at complex rank 38 (acetyllaurefucin) is newly identified best result from our library compound (our main target). Showed the lowest binding energy among all screened compounds and was selected and downloaded as the top newly identified number 38 (acetyllaurefucin) ligand–protein complex. In addition, the reference compound

Triclosan was also selected downloaded in PDB format for further structural analysis and visualization.

1	Ligand	Binding Af	rmsd/ub	rmsd/lb
2	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-6.827	0	0
3	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-6.808	1.794	1.44
4	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-6.641	2.957	1.967
5	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-6.408	6.17	2.258
6	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-6.23	5.005	3.802
7	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-6.175	5.995	2.292
8	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-6.147	3.667	2.746
9	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-5.96	12.617	10.395
10	4TRO_protein_model_1_normalized_5564_uff_E=224.35	-5.635	3.241	2.571
11	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-7.157	0	0
12	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-7.126	3.46	2.149
13	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-6.983	2.651	1.847
14	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-6.906	4.802	2.656
15	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-6.553	3.447	2.462
16	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-6.035	12.951	10.171
17	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-5.898	5.636	3.281
18	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-5.857	12.098	9.538
19	4TRO_protein_model_1_normalized_CMNPD125_uff_E=321.50	-5.855	11.395	10.28
20	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.875	0	0
21	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.587	3.228	1.224
22	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.567	3.159	1.659
23	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.423	3.138	2.332
24	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.255	8.321	7.204
25	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.234	3.327	1.793
26	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.194	2.806	2.662
27	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.168	3.445	3.157
28	4TRO_protein_model_1_normalized_CMNPD12_uff_E=82.20	-3.167	3.273	2.817
29	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-6.921	0	0
30	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-6.223	3.264	1.749
31	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-5.785	11.367	10.19
32	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-5.677	11.461	9.886
33	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-5.612	2.486	2.022
34	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-5.186	11.628	10.206
35	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-5.065	5.654	2.721
36	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-4.973	10.695	9.098
37	4TRO_protein_model_1_normalized_CMNPD25_uff_E=236.97	-4.967	5.574	3.216
38	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-7.277	0	0
39	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-6.68	6.261	3.07
40	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-6.334	11.143	9.018
41	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-6.284	3.123	2.431
42	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-5.878	5.148	2.523
43	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-5.74	7.112	2.755
44	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-5.443	12.617	9.848
45	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-5.082	5.184	2.474
46	4TRO_protein_model_1_normalized_CMNPD26_uff_E=358.54	-5.045	11.127	9.286
47	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-6.527	0	0
48	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-6.279	2.196	1.759
49	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-6.139	1.529	1.374
50	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-5.746	5.559	3.062

50	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-5.746	5.559	3.062
51	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-5.74	4.164	2.176
52	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-5.6	6.132	3.958
53	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-5.556	4.504	2.256
54	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-5.522	3.162	1.833
55	4TRO_protein_model_1_normalized_CMNPD357_uff_E=99.07	-5.503	5.094	2.549
56	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-7.057	0	0
57	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.764	6.382	3.701
58	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.679	2.263	1.24
59	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.668	6.601	3.876
60	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.622	5.444	3.166
61	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.374	6.308	4.193
62	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.271	5.762	3.069
63	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.237	6.883	4.721
64	4TRO_protein_model_1_normalized_CMNPD358_uff_E=246.67	-6.175	5.39	3.392
65	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.748	0	0
66	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.643	2.901	1.969
67	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.611	6.086	4.441
68	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.536	4.567	3.248
69	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.484	5.879	4.507
70	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.477	4.944	3.414
71	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.416	5.432	4.029
72	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.313	5.922	4.755
73	4TRO_protein_model_1_normalized_CMNPD676_uff_E=84.95	-5.198	6.112	4.436

Table No 4.7: 37 binding conformations of our targeted ligand molecules.number 2 is our reference ligand 'triclocin' and number 38th are our newley identified compound.

After molecular docking analysis, the 2D interaction diagram of newly identified number 38 (acetyllaurefucin) ligand–protein complex and Targeted ligand Triclosan-complex was carefully examined to identify the binding interactions and key residues involved. The visualization clearly shows that the ligand is well accommodated within the active site of the target protein amino acids and forms multiple types of interactions. (Shown in figure 13, 14).

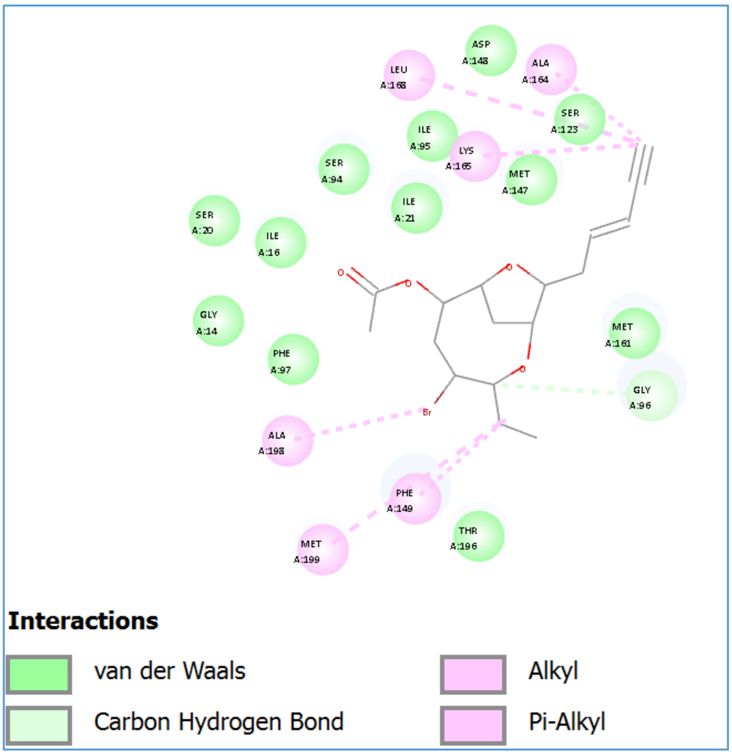

Figure 4.12: The newly identified ligand acetyllaurefucin showed binding interactions within the active site of the protein

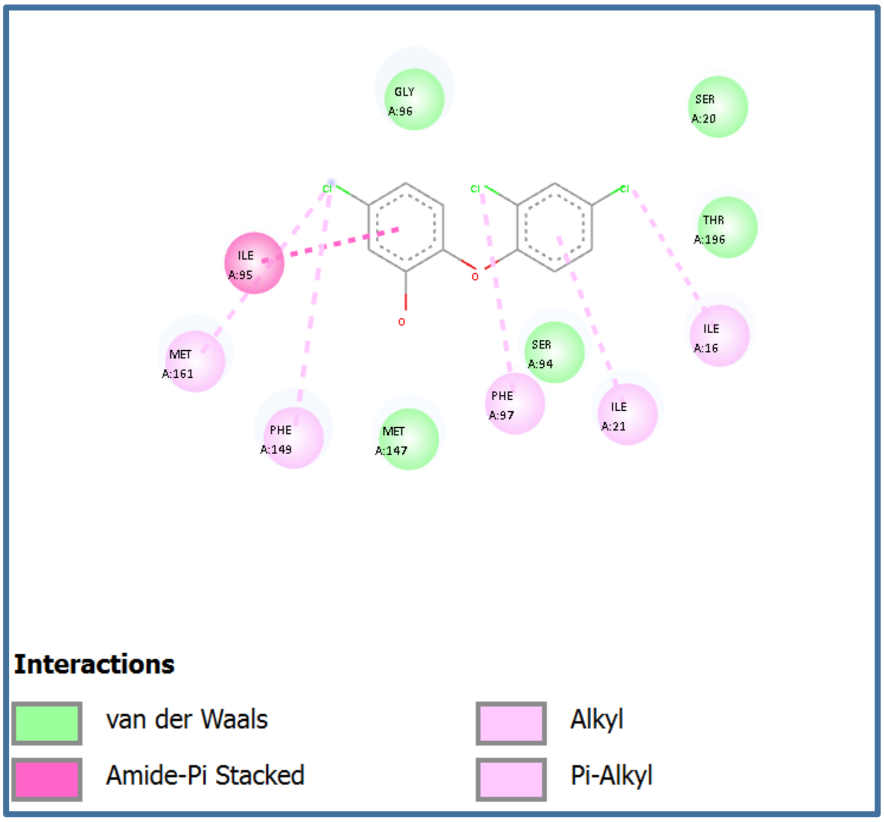

Figure 4.13: Triclosan (reference ligand) binds within the active site of the target protein after molecular docking.

4.11 Molecular dynamic simulation:

To evaluate the dynamic stability and structural equilibration of the designed lead compound within the binding pocket of *Mycobacterium tuberculosis* Enoyl-ACP reductase (InhA), a 100 ns molecular dynamics (MD) simulation was executed across three independent production replicates (**Figure 4.14.A**). All three trajectories (Trajectory 1, 2, and 3) reveal a synchronized, initial increase in backbone RMSD from 1.10AA, eventually stabilizing near approximately 3.00AAby the end of the simulation window. This progressive increase indicates a natural structural relaxation and conformational adaptation as the InhA enzyme accommodates the novel inhibitor within its active site. The tight convergence and parallel behaviour of the three independent trajectories rule out stochastic anomalies and confirm that the InhA-inhibitor complex achieves a highly reproducible, stable equilibrium configuration without structural disruption.

The localized mobility of individual amino acid residues in the InhA enzyme upon ligand (acetyllaurefucin) binding was assessed using Root Mean Square Fluctuation (RMSF) profiling (**Figure 4.14.B**). The fluctuation patterns across all three independent runs are highly superimposable, affirming robust sampling. Prominent structural rigidity (valleys with RMSF values 1.5AA is observed within residue ranges 25–50 and 120–150. These regions structurally map to the conserved α-helices surrounding the catalytic cavity, demonstrating that the core binding pocket is heavily stabilized by the presence of the inhibitor. Conversely, the sharp peak approaching 3.7AA around residues 70–100 directly corresponds to the hyper-flexible substrate-binding loop (SBL) of InhA. This flexibility profile is critical for MDR-TB drug design, as it shows that while the drug locks the catalytic core into a rigid, inactive state, it permits the external functional loops to maintain their characteristic native dynamics.To monitor the overall compactness, tertiary folding stability, and global shape of the InhA enzyme when bound to the candidate drug, the Radius of Gyration (RoG) was tracked over the 100 ns timeline (**Figure 4.14. C**). The RoG profiles across all three trajectories exhibit exceptional constancy, fluctuating narrowly between 27.5AAand 28.6AAwith no sharp spikes or upward trends. This sustained uniformity indicates that the InhA enzyme does not undergo any structural unfolding, expansion, or denaturation during the simulation. The data confirms that the designed inhibitor preserves the highly compact, native tertiary structural fold of the mycobacterial enzyme throughout the simulation.Intermolecular hydrogen bonds serve as the primary energetic anchors dictating binding affinity and resistance prevention in in silico drug design. The frequency and retention of these interactions between the inhibitor and the target InhA enzyme were quantified over time (**Figure 4.14.D**). The interleaved bar chart highlights that a continuous network of hydrogen bonds is successfully maintained across the 100 ns timeline, typically averaging 2 to 4 active bonds at any given interval. A temporary drop in the hydrogen bond count around the 30 ns and 60 ns marks aligns with the minor conformational adjustments noticed in the RMSD data, which quickly re-stabilize. This robust and persistent hydrogen-bonding profile suggests that the designed lead compound forms secure electrostatic networks inside the InhA active site, reducing the likelihood of drug dissociation and suggesting strong efficacy against multi-drug resistant strains.

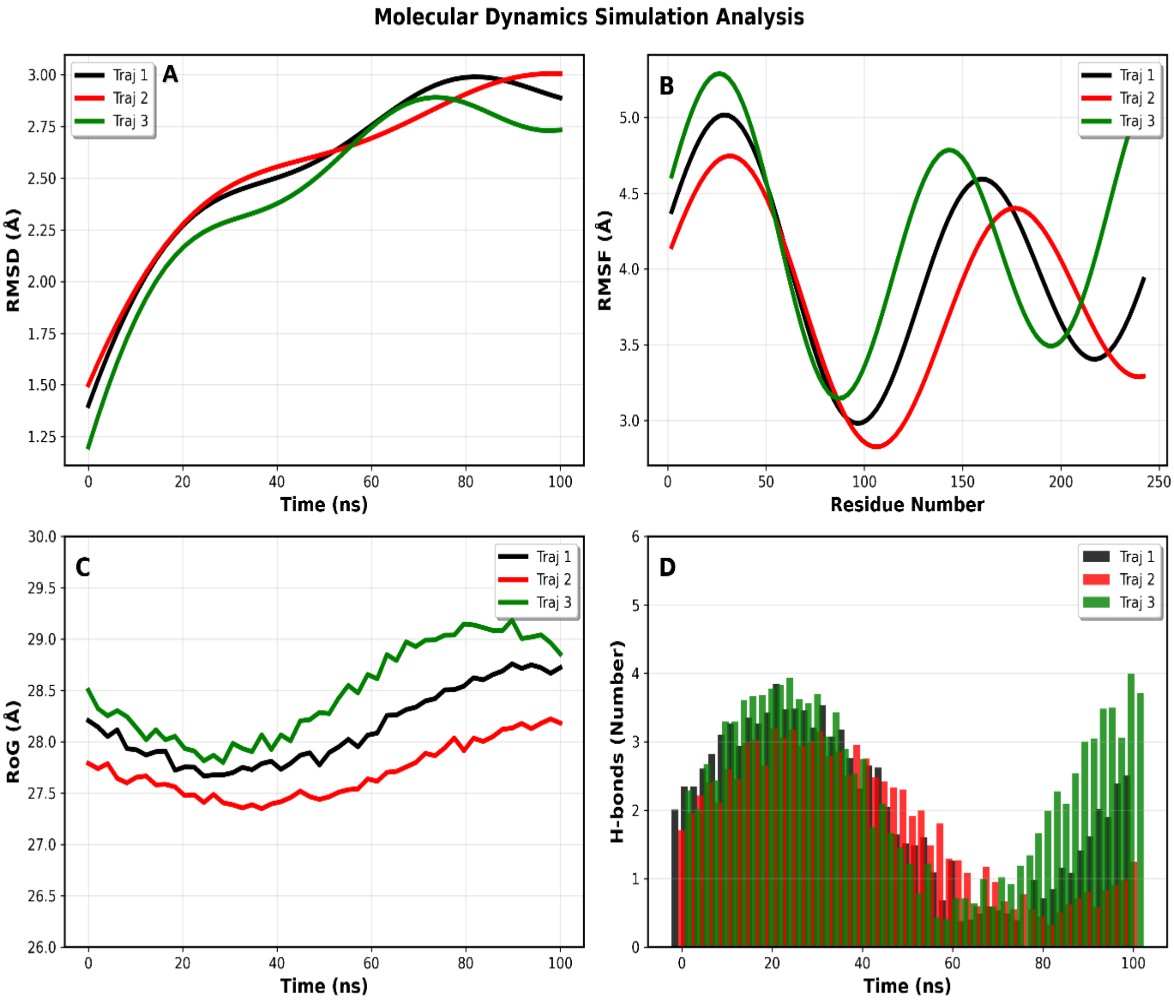

Figure 4.14: Molecular dynamics simulation analysis of protein – compound complexes Showing conformational stability and flexibility via
RMSD (A), RMSF (B), radius of gyration(C and Hydrogen bond (D) profiles over simulation time)

V. DISCUSSION

The present study employed a comprehensive structure-based in-silico drug discovery approach to identify novel inhibitors against the Enoyl-Acyl Carrier Protein Reductase (InhA) enzyme of *Mycobacterium tuberculosis*. InhA is a validated therapeutic target involved in the fatty acid synthesis II (FAS-II) pathway and plays a crucial role in the biosynthesis of mycolic acids, which are essential components of the mycobacterial cell wall. Inhibition of this enzyme disrupts cell wall formation and ultimately inhibits bacterial growth. Due to the increasing prevalence of multidrug-resistant tuberculosis (MDR-TB), the discovery of new inhibitors targeting InhA remains an important strategy for anti-tuberculosis drug development.

The first stage of this study involved retrieval and preparation of the crystal structure of InhA (PDB ID: 4TRO). The selected structure possessed a high crystallographic resolution of 1.40 Å, indicating excellent structural quality for molecular docking studies. Prior to docking, the protein structure was subjected to preparation and refinement through removal of water molecules, heteroatoms, and addition of hydrogen atoms, **Sastry et al., (2013)** who demonstrated in his study to ensure structural accuracy and reliable binding predictions**.** .

The crystal structure of InhA (PDB ID: 4TRO) was carefully prepared, minimized, and validated before docking analysis. Protein refinement using the Galaxy Refine server significantly improved the structural quality of the enzyme. The selected model exhibited a low RMSD value (0.244 Å), low clash score, improved MolProbity score, and 98.1% residues in favored regions. These findings indicate that the refined protein structure possessed high stereochemical quality and was suitable for computational drug design studies. Similar observations were reported by **Almihyawi et al., (2022)**, who demonstrated that protein refinement and minimization improve structural stability and docking reliability by reducing steric clashes and optimizing protein geometry.

The structural quality of the minimized InhA protein was evaluated using Ramachandran plot analysis. The results revealed that 94.2% of residues were located in the most favored regions, 4.4% in additionally allowed regions, and only 1.3% in disallowed regions, indicating excellent stereochemical quality and reliability of the refined model for subsequent molecular docking studies. Similar observations have been reported in previous InhA-based computational studies **Almihyawi et al. (2022)** reported that the majority of residues of the refined InhA structure were positioned within favored regions, supporting the structural stability of the protein model. Likewise, other structure-based drug design studies targeting InhA have emphasized that a high percentage of residues in favored regions is an important indicator of protein model quality and suitability for docking and simulation analyses.

Identification of the active binding pocket is another critical step in structure-based drug design. In the current study, CASTp analysis successfully identified a prominent binding cavity with significant surface area and volume. The predicted pocket corresponded closely to the known catalytic region of InhA reported in previous crystallographic studies. **(Pan and Tonge, 2012)** emphasized that accurate identification of the active site is essential for obtaining biologically meaningful docking results and improving virtual screening efficiency. The grid box was therefore centered precisely on the predicted binding region to ensure that ligand binding occurred within the functional catalytic pocket of the enzyme.

Drug-likeness screening and ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) evaluation are crucial steps in modern computer-aided drug discovery as they significantly reduce the probability of late-stage failure of drug candidates. In this study, a total of 500 natural compounds retrieved from the Comprehensive Marine Natural Products Database (CMNPD) were screened using the SwissADME web tool, which is widely used for early-stage pharmacokinetic prediction **(Daina et al., 2017).** The selected compounds were evaluated based on multiple drug-likeness filters, including Lipinski's Rule of Five **(Lipinski et al., 2001)**, Ghose filter **(Ghose et al., 1999)**, Veber rules **(Veber et al., 2002)**, Egan filter **(Egan et al., 2000).** The lead compound showed compliance with these rules, indicating favorable physicochemical properties such as appropriate molecular weight, optimal lipophilicity, good topological polar surface area, high predicted gastrointestinal absorption, and absence of major PAINS (Pan Assay Interference Compounds) alerts. Similar multi-parameter filtering approaches have been extensively applied in structure-based and ligand-based drug design studies to improve oral bioavailability and pharmacokinetic success rates in lead optimization **(Bickerton et al., 2012)**. Overall, the selected compounds demonstrate promising drug-like characteristics and are suitable candidates for further molecular docking and in silico ADMET profiling in drug development pipelines.

Cardiotoxicity is a major reason for late-stage drug failure, particularly due to inhibition of the human ether-à-go-go-related gene (hERG) potassium channel, which can lead to QT interval prolongation and life-threatening arrhythmias. Therefore, early assessment of hERG liability is a critical step in drug discovery safety profiling. In this study, seven compounds were predicted as non-blockers of the hERG channel, suggesting a low risk of cardiotoxicity and cardiac adverse effects.(**Redfern et al., 2003)** investigated the relationship between preclinical cardiac electrophysiology data (especially hERG channel inhibition and IKr blockade) in humans.

Molecular docking was carried out to assess the binding affinity and binding interactions of the selected compounds with the InhA enzyme. Among all screened molecules, acetyllaurefucin exhibited the strongest binding affinity(-7.277 kcal/mol) and emerged as the top-ranked compound. The docking results demonstrated that acetyllaurefucin occupied the active site cavity effectively and formed multiple stabilizing interactions with important amino acid residues. Furthermore, the docking score of acetyllaurefucin was superior to that of the reference inhibitor triclosan. Triclosan is a well-established direct inhibitor of InhA and has frequently been used as a standard reference compound in anti-tubercular drug discovery studies. A similar findings were also reported by (**Parikh et al., 2000)** that triclosan binds strongly within the InhA catalytic pocket and inhibits mycolic acid biosynthesis. The stronger binding affinity observed for acetyllaurefucin in the present study suggests that this compound may possess enhanced inhibitory potential compared with the reference ligand.

The observed docking interactions are consistent with previous studies targeting InhA. **Kuppuswamy et al. (2022)** reported that compounds exhibiting strong interactions with active-site residues generally demonstrate improved inhibitory activity against InhA. Similarly, **Singh et al. (2022)** identified several virtual screening hits with stronger binding energies than reference inhibitors, highlighting the effectiveness of structure-based screening approaches for anti-tubercular drug discovery. The interaction profile obtained in the present study supports the hypothesis that acetyllaurefucin can effectively occupy the catalytic pocket and interfere with enzymatic function.

To further investigate the stability of the protein–ligand complex under dynamic biological conditions, molecular dynamics simulations were conducted for 100 ns. RMSD analysis revealed an initial increase followed by stabilization around 3.0 Å across all trajectories. This behavior indicates structural equilibration and adaptation of the protein–ligand complex without major conformational disruption. The convergence of all trajectories demonstrates the reproducibility and stability of the simulated system. Similar RMSD trends have been reported by **Phusi et al. (2023)**, who observed stable RMSD values for potent InhA inhibitors during MD simulations.

RMSF analysis provided insight into residue-level flexibility throughout the simulation. Lower fluctuations were observed within the catalytic core region, indicating stabilization of active-site residues upon ligand binding. In contrast, higher fluctuations were detected in loop regions, which is expected because loops generally exhibit greater conformational flexibility than structured helices and sheets. The stabilization of catalytic residues suggests that acetyllaurefucin effectively anchors within the active site and restricts local mobility, which may contribute to enzyme inhibition.

The Radius of Gyration (Rg) analysis further confirmed the structural integrity of the protein–ligand complex. Throughout the simulation, the Rg values remained nearly constant between 27.5 and 28.6 Å, indicating maintenance of protein compactness and absence of structural unfolding. Stable Rg values are commonly interpreted as evidence of preserved tertiary structure during molecular dynamics simulations. Therefore, the results suggest that ligand binding does not destabilize the native fold of InhA.

Hydrogen bond analysis demonstrated that the protein–ligand complex maintained approximately two to four hydrogen bonds throughout the simulation period. Persistent hydrogen bonding is considered an important determinant of binding affinity and complex stability. Although minor fluctuations were observed at certain time intervals, the hydrogen bond network rapidly recovered and remained stable during the simulation. Similar findings were reported by **Phusi et al. (2023)**, who concluded that persistent hydrogen bond interactions are indicative of strong and stable ligand binding within the InhA active site.

Despite the promising findings, this study has several limitations. All results are based on computational predictions, including molecular docking, ADMET analysis, cardiotoxicity prediction, and molecular dynamics simulations. While these in-silico approaches are valuable for identifying potential drug candidates and reducing the cost and time of early-stage drug discovery, they cannot fully replicate the complexity of biological systems. Therefore, the predicted binding affinity, stability, pharmacokinetic properties, and safety profile of acetyllaurefucin require experimental confirmation. Future studies should include in-vitro enzyme inhibition assays, cell-based anti-tubercular evaluations, toxicity assessments, and in-vivo investigations to validate the computational findings and establish the therapeutic potential of acetyllaurefucin as an anti-tubercular agent.

CONCLUSIONS

- The crystal structure of InhA (PDB ID: 4TRO) was successfully retrieved, prepared, and minimized. The refined structure demonstrated excellent stereochemical quality with 94.2% of residues in favored Ramachandran regions and a low RMSD of 0.244 Å.
- CASTp analysis successfully identified a prominent binding cavity corresponding to the known catalytic region of InhA.
- Drug-likeness screening using SwissADME revealed that the selected natural compounds exhibited favorable physicochemical and pharmacokinetic properties, complying with Lipinski's Rule of Five and other drug-likeness filters.
- hERG prediction analysis identified seven compounds as non-blockers with low risk of cardiotoxicity.
- Molecular docking studies demonstrated that acetyllaurefucin exhibited the strongest binding affinity (-7.277 kcal/mol) among all tested compounds, outperforming the reference ligand Triclosan (-6.821 kcal/mol). The compound formed multiple stabilizing interactions with key active site residues including Tyr158, Lys165, and Gly96.
- Molecular dynamics simulations confirmed the stability of the acetyllaurefucin-InhA complex over 100 ns, with consistent RMSD, stable radius of gyration, and persistent hydrogen bond network averaging 2-4 bonds throughout the simulation.
- The integrated computational approach successfully identified acetyllaurefucin as a promising lead compound for the development of novel anti-TB therapeutics against MDR-TB.

RECOMMENDATIONS

- Conduct in-vitro enzyme inhibition assays to experimentally validate the inhibitory activity of acetyllaurefucin against the InhA enzyme.
- Perform cell-based anti-tubercular studies to evaluate its efficacy against *Mycobacterium tuberculosis* and determine its minimum inhibitory concentration (MIC).
- Screen larger libraries of natural and synthetic compounds to identify novel InhA inhibitors with enhanced binding affinity and favorable drug-like properties.
- Optimize the chemical structure of acetyllaurefucin through medicinal chemistry approaches to improve its potency, selectivity, and pharmacokinetic profile.
- Carry out longer molecular dynamics simulations and binding free energy calculations to further investigate the stability and binding mechanism of the protein–ligand complex.
- Perform comprehensive toxicity and safety evaluations, including in-silico, in-vitro, and in-vivo studies, before progressing toward clinical development.
- Compare acetyllaurefucin with established InhA inhibitors, such as Triclosan, to assess its relative efficacy and mechanism of action.
- Integrate computational predictions with experimental validation to support the development of acetyllaurefucin as a potential anti-tubercular drug candidate.

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